AP CALCULUS AB

Unit 1 • Lesson 1.5 practice • Answer key

Lesson 1.5 • Determining Limits Using Algebraic Properties of Limits

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

$\lim_{x \to 2} \left(3x^2 - x\right)$ is

(A) 10

(B) 12

(C) 14

(D) 8

Solution

Answer: (A). Direct substitution: $3(4) - 2 = 10$.

Question 2

If $\lim_{x \to c} f(x) = 4$ and $\lim_{x \to c} g(x) = -1$, then $\lim_{x \to c} \left(f(x)g(x) + 3\right)$ is

(A) 7

(B) -1

(C) 3

(D) -4

Solution

Answer: (B). $4 \cdot (-1) + 3 = -1$.

Question 3

If $\lim_{x \to c} f(x) = 6$ and $\lim_{x \to c} g(x) = 2$, then $\lim_{x \to c} \dfrac{f(x)}{g(x)}$ is

(A) 12

(B) 4

(C) 3

(D) $\dfrac{1}{3}$

Solution

Answer: (C). The denominator's limit is not zero, so the limit is $\frac{6}{2} = 3$.

Question 4

$\lim_{x \to \pi} \cos x$ is

(A) 0

(B) 1

(C) π

(D) -1

Solution

Answer: (D). $\cos x$ is continuous, so the limit is $\cos \pi = -1$.

Question 5 (Free response)

$\lim_{x \to 1} f(x) = 3$, $\lim_{x \to 1} g(x) = -2$ and $h(x) = x^2 + 1$. Find each limit, showing which properties you use.

(a) $\lim_{x \to 1} \left(2f(x) - g(x)\right)$

(b) $\lim_{x \to 1} f(x)h(x)$

(c) $\lim_{x \to 1} \sqrt{f(x) + 6}$ (d) $\lim_{x \to 1} \big(g(x)\big)^3$

Solution

(a) Constant multiple and difference: $2(3) - (-2) = 8$.
(b) $\lim_{x \to 1} h(x) = 2$ by substitution; product: $3 \cdot 2 = 6$.
(c) Sum, then the square root (continuous): $\sqrt{3 + 6} = 3$.
(d) Power: $(-2)^3 = -8$.